



Research Paper

Mental health among ethnic minority adolescents in Vietnam and correlated factors: A cross-sectional study

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ABSTRACT

Objective: This study aimed to measure the rate of depression, anxiety and stress among ethnic minority adolescents and identify relationships between demographic characteristics, behaviours and adverse childhood experiences (ACEs) factors with different mental disorders.

Methods: A cross-sectional study was conducted in Lang Son province, Vietnam, among adolescents in boarding schools for ethnic minorities. The Depression-Anxiety-Stress Scale-21 (DASS-21) and the ACEs Questionnaire were used. Information on age, gender, ethnicity, frequency of home visits, frequency of parental visits, number of people living together, quality of friendship, physical activity, and frequency of Internet use was also collected. Multivariate Logistic and Tobit regression were employed.

Results: Among 845 students, the rates of stress, anxiety and depression were 24.7 %, 54.4 % and 59.0 %, respectively. There were 48.9 % experiencing at least one ACEs, and the mean number of ACEs was 1.1 (SD = 1.8, range = 0–6). The number of ACEs was positively associated with anxiety score (Coef. = 0.28), stress score (Coef. = 0.28), and depression score (Coef. = 0.16). Having fair or poor friendships was positively associated with anxiety, stress and depression scores. Adolescents using the Internet daily or not using the Internet had a significantly lower anxiety score than those using the Internet at the weekend. Finally, information sources such as friends or television were negatively associated with stress and anxiety scores.

Conclusion: This study revealed the high prevalence of mental health problems and ACEs among ethnic minority adolescents in current sample. The present study illustrates an immediate demand for interventions to address ACEs and cognitive issues to improve adolescents' well-being.

1. Introduction

In recent years, there has been a global increase in the prevalence of mental health disorders among adolescents (World-Health-Organization 2021). This is now a common health concern affecting many young people worldwide. Mental health issues make up 13 % of the overall burden of disease and injury worldwide among individuals aged 10–19 years, with the majority of cases going unnoticed and untreated (World-Health-Organization 2021). Depression is a predominant contributor to illness, disability, and mortality among teenagers globally. In addition to undergoing a range of physiological, psychological, and sociological transformations, being subjected to adverse childhood experiences (ACEs) has been recognised as a significant causative element in the development of mental health issues among adolescents

(Prevention CfDCA 2023; Thai et al., 2020; Herzog and Schmahl, 2018). Previous studies found that problems in relationships with peers (Aldam et al., 2019) or risk behaviours such as alcohol use, smoking (Veisani et al., 2020) or excessive Internet use (Veisani et al., 2020) increased the risk of mental problems in adolescents.

ACEs encompass traumatic occurrences with various forms of maltreatment such as abuse, neglect, and violence, which manifest in individuals below the age of 18 (Prevention CfDCA 2023). The experience of ACEs is associated with an increased probability of encountering complications connected to traumatic stress during adulthood (Barboza, 2018). Such complications encompass inadequate health behaviours, mental health disorders, and long-standing health conditions. An examination in the United States encompassing a sample of 933 adolescents aged 14 years revealed that over 90 % of the participants

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experienced at least one type of ACEs, with approximately 57 % reporting three or more ACEs (Flaherty et al., 2013). A separate investigation conducted on a sample size of 14,661 individuals from the younger age group across 10 European nations displayed that close to 50 % of the participants encountered at least one Adverse Childhood Experience (ACE). In contrast, approximately 25 % of them reported the occurrence of multiple ACEs (Hughes et al., 2019). Asian countries have also disclosed high rates of violence committed against children, including child sexual abuse, physical abuse, and emotional abuse (Fang et al., 2015).

The occurrence of ACEs during the early stages of life engenders immediate as well as enduring repercussions on the progression of mental health throughout an individual's lifetime (Desch et al., 2023). ACEs are linked with psychosocial and economic deficits, a noteworthy decline in quality of life, engagement in risky behaviour, and an elevated prevalence of both mental and physical health issues (Thai et al., 2020; Hughes et al., 2019; Desch et al., 2023). A correlation has been established between childhood adversity and reduced social support, resulting in adverse effects on mental health. This reduced social support acts as a mediator in the relationship between ACEs and mental health outcomes (Karatekin and Ahluwalia, 2020). Racial/ethnic minority children were more likely to suffer mental problems after adjusting to ACEs (Zarei et al., 2023; LaBrenz et al., 2020). Moreover, ACEs are correlated with decreased household income levels, diminished financial well-being and a lack of financial literacy (Harter and Harter, 2022). The COVID-19 pandemic even amplifies the risk of mental disorders among children with ACEs (Clemens et al., 2022).

In the context of Vietnam, there has been a significant occurrence of ACEs and mental health issues among adolescents (Tran et al., 2017). In 2014, UNICEF reported that approximately 68.4 % of Vietnamese children between the ages of 1 and 14 encountered one or more instances of physical or psychological discipline from their parents or caregivers (UNICEF 2017). A prior study in the northern region of Vietnam, involving a sample of 1851 adolescents aged between 12 and 17 years, indicated that close to fifty per cent of the sampled students had experienced one or more ACEs within the previous year (Tran et al., 2017). Meanwhile, according to a recent epidemiological study conducted across ten provinces, the prevalence of mental disorders in children was approximately 12 %, which implies that nearly 3 million children require mental health interventions (Weiss et al., 2014). However, most of the studies were performed in urban or rural settings, and there remains a dearth of scholarly inquiry on the analysis of these phenomena among ethnic minority adolescents in Vietnam, particularly among adolescents studying in boarding schools.

In Vietnam, adolescents of ethnic minorities who study in boarding schools possess distinct psychological characteristics compared to the general high school student population (Quyen, 2014). Various outdated cultural practices have significantly impacted this group. The individual prefers to live in a state of freedom and autonomy without being constrained by societal norms and collective regulations. The majority of adolescents exhibit psychological tendencies towards self-consciousness, inadequacy, or self-centeredness. However, in addition to that, they are also very sincere, honest, and able to endure difficulties and hardships (Quyen, 2014). The youths have a profound awareness of human beings and life, a responsibility to the community, and a correct perception of the development of the economy, society, and culture that impact their lives, families, and hometowns. The societal and economic changes may give rise to certain negative psychological tendencies. For example, in psychology and selfishness, some individuals view themselves as entitled to priority in all situations, policies, and governance (Quyen, 2014). The self-perception of minority ethnic students tends to be more natural, and their cognitive and emotional development is relatively well-developed, including sensory and kinetic perception. However, this may result in a lack of comprehensive and critical thinking. In conjunction with the aforementioned characteristics, the educational pursuits of minority students are further

impeded by an ingrained rigidity. The ability to analyze, speculate, and engage in logical reasoning is limited. The education issue needs to be prioritised due to insufficient incentives for advancement. Thus, in addition to various psychological characteristics resulting from cultural differences, living environment, and family circumstances.

To provide more empirical evidence for further mental health interventions in ethnic minority children and adolescents, this study aimed to measure the rate of depression, anxiety and stress among ethnic minority adolescents who studied in boarding schools within a mountainous province, and identify relationships between demographic characteristics, behaviors and ACEs with different mental disorders.

2. Materials and methods

2.1. Study settings

A cross-sectional study was conducted in Lang Son province from August to December 2022. Lang Son is a mountainous province located in the Northeast region of Vietnam, with a population of 781,655 people (according to the population census on April 1, 2019). The province is inhabited by seven main ethnic groups, namely Nung comprising 42.97 % of the population, Tay comprising 35.92 %, Kinh comprising 16.5 %, while the remaining percentage consists of other ethnic groups such as Dao, Hoa, San Chay, and Hmong with a combined total of 4.61 %. The urban population comprises 20.40 % of the total population, while the rural population constitutes 79.60 %. The study participants comprised students enrolled at boarding secondary and high schools for ethnic minorities within the province of Lang Son, ranging in age from 10 to 18, who voluntarily consented to participate in the research. The exclusion criteria pertain to students who refused to participate in the research or failed to complete the survey. We randomly selected boarding secondary and high schools within the province of Lang Son. The boarding school is a type of school that is only for ethnic minority adolescents in mountainous provinces of Vietnam. In the selected schools for our study, we encompassed the entire student population across all grade levels from grades 6 to 12. A total of 845 students participated in the research study. The research was approved by the Provincial People's Committee of Lang Son.

2.2. Data collection

In this study, a structured questionnaire was developed for data collection. The set of questions encompassed demographic characteristics, behaviors, relationship characteristics, the Depression-Anxiety-Stress Scale – 21 (DASS-21) instrument, and the ACEs toolkit. The questionnaire was tested on 10 adolescents who met the selection criteria and subsequently modified to ensure logical sequencing and appropriate language usage. The final version was approved by the research group and leaders of selected schools.

The investigator received training on the objectives and benefits of the study, the content of the research questions, and the information on the data collection tool. Subsequently, the research team proceeded to obtain institutional approval for the research implementation and presented the research content to the teachers. The teacher guided students in requesting permission from their parents to participate in the research. The investigator introduced the content of the study and provided specific instructions on how to fill in the information. The investigator directly elucidated the contents that students had inquiries about and oversaw the process of students completing survey information. Once the evaluation forms were collected, the investigator proceeded to verify the completeness and appropriateness of the information provided in the forms. If any information was lacking or unsuitable, immediate amendments were made by requesting the student provide additional details.

2.2.1. Demographic characteristics, behaviors and relationship characteristics

We obtained information about age (years old), gender (male/female), ethnicity (Tay/Nung/others), frequency of home visits (Once a month/Every two months/Every three months or less), frequency of parental visits (Once a month/Every two months/Others), number of people living together (<10 people/ ≥ 10 people), quality of friendship (Good/Fair/Poor), physical activity (Yes/No), and frequency of Internet use (Weekend/4 h per day/5 h per day/ 6 h per day or more).

2.2.2. Mental health

The DASS 21 scoring system consists of 21 questions aimed at evaluating the domains of depression, anxiety, and stress (Le et al., 2017). In each domain, there are seven questions for children to personally fill out to express their subjective experiences of life over the past two weeks, ranging from levels of normal, mild, moderate, severe, and very severe. The DASS 21 not only allows for the assessment of the severity levels in patients but also serves as a method to understand the patient's response to treatment (Le et al., 2017). The DASS 21 questionnaire consists of a series of questions, each utilizing a Likert scale ranging from 0 to 3, which is encoded as follows: Never (0 points), Sometimes (1 point), Often (2 points), and Almost Always (3 points). The total scores ranged from 0 to 21, with a higher score indicating a higher level of mental disorders. For depression, children were assigned into five categories: normal (0–4), mild (5–6), moderate (7–10), severe (11–13) and extremely severe (14+). For anxiety, children were assigned into five categories: normal (0–3), mild (4–5), moderate (6–7), severe (8–9) and highly severe (10+). For stress, adolescents were assigned into five categories: normal (0–7), mild (8–9), moderate (10–12), severe (13–16) and highly severe (17+). These classifications followed the guidelines in the prior study (Henry and Crawford, 2005). The Vietnamese instrument showed good validity and reliability among adolescents (Le et al., 2017). The Cronbach's alpha for the Depression, Anxiety and Stress subscales were 0.767; 0.792 and 0.811, respectively. Moreover, we asked participants to report the information sources for mental health (Family/School/Friends/Commune staff/Youth Union/Women Union/Television/Self-study), as well as the coping strategies (Reading books/Listening to music/Watching television/Playing sport/Sleeping/Talking/Crying/Walking) if they met any mental problems.

2.2.3. Adverse childhood experiences

The Adverse Childhood Experiences Questionnaire (ACE-Q) was used to measure the ACEs among adolescents (Felitti et al., 1998). It is a standardized assessment tool comprising ten items, designed to quantitatively capture the occurrence of adverse or traumatic experiences that individuals have encountered in their early stages of life (before reaching the age of 18). The ACE-Q assessment instrument is designed to evaluate the extent of a child's exposure to ACEs, encompassing psychological, physical, and sexual abuse, alongside household dysfunction such as domestic violence, substance use, and incarceration. Each question had two options for their responses: Yes=1, No=0. The aggregate ACE-Q score (ranging from 0 to 10) is determined by summing the affirmative answers to all questions. A higher ACE-Q score indicates more ACEs than the participants experienced (Felitti et al., 1998). The Cronbach's alpha of this instrument was 0.629.

2.3. Statistical analysis

Stata software version 15.0 was used for data analysis. Descriptive statistics were performed. Multivariate Logistic and Tobit regression were employed to measure associations between ACEs and mental disorders, as well as DASS-21 scores. A stepwise forward strategy was used to build reduced models, with a p-value of log-likelihood of <0.2 . Statistical significance was defined if a p-value was less than 0.05.

3. Results

Among 845 students, the mean age was 14.6 (SD=1.6). Most of them were female (70.7 %) and belonged to the Nung ethnicity (50.4 %). The majority of them returned home once a month (91.0 %), and 87.8 % had parental visits once a month. Most of the students lived in the dormitory with less than 10 people (61.4 %). The highest proportion of students had good friendships (54.1 %). Most of them performed physical activity in their free time (67.9 %), and used the Internet at the weekend (60.7 %) (Table 1).

The mean number of ACEs was 1.1 (SD=1.8, range=0–6). The prevalence of children experiencing emotional abuse was the highest (21.9 %), followed by emotional neglect (14.4 %) and parental divorce (11.0 %). The rates of sexual abuse (0.9 %) and physical neglect (0.3 %) were the lowest. In general, 48.7 % of students experienced at least one ACE, and 13.8 % experienced three or more such experiences (Table 2).

Table 3 shows that the rates of stress, anxiety and depression were 24.7 %, 54.4 % and 59.0 %, respectively. The most common strategies to cope with mental problems were reading books (41.8 %), listening to music (12.8 %) and sleeping (10.8 %). The main information sources were from schools (35.5 %), self-study (18.6 %) and television (15.2 %).

Fig. 1 shows that the mean number of ACEs increased along with the higher levels of stress, anxiety and depression ($p < 0.05$).

Table 4 shows that the number of ACEs was positively associated with anxiety score (Coef.=0.28), stress score (Coef.=0.28), and depression score (Coef.=0.16). In addition, each ACE increased 1.2 times the odds of having anxiety (OR=1.20) and 1.2 times the odds of having stress (OR=1.20).

Regarding other factors, living with more than ten people was negatively associated with the odds of having anxiety (OR=0.51) and negatively correlated with anxiety score (Coef. = -0.87). Adolescents doing physical activity were less likely to have anxiety (OR=0.72). Having fair or poor friendships was positively associated with anxiety, stress and depression scores. Adolescents using the Internet daily or not

Table 1
Demographic characteristics, behaviors, and relationship characteristics among students.

Characteristics	Values
Gender, n(%)	248 (29.3 %)
	Female 597 (70.7 %)
Ethnicity, n(%)	Tay 326 (38.6 %)
	Nung 426 (50.4 %)
	Others 93 (11.0 %)
Frequency of home visit, n(%)	Once a month 769 (91.0 %)
	Every two months 58 (6.9 %)
	Every three months or less 18 (2.1 %)
Frequency of parental visit, n(%)	Once a month 742 (87.8 %)
	Every two months 85 (10.1 %)
	Others 18 (2.1 %)
Number of people living together, n (%)	< 10 people 519 (61.4 %)
	≥ 10 people 326 (38.6 %)
Quality of friendship, n(%)	Good 457 (54.1 %)
	Fair 360 (42.6 %)
	Poor 28 (3.3 %)
Physical activity, n(%)	Yes 574 (67.9 %)
	No 271 (32.1 %)
Frequency of Internet use, n(%)	No use 25 (3.0 %)
	Weekend 513 (60.7 %)
	4 h/day 204 (24.1 %)
	5 h/day 53 (6.3 %)
	6 h/day or more 50 (5.9 %)
Number of ACEs	0 ACE 434 (51.3 %)
	1 ACE 192 (22.7 %)
	2 ACEs 103 (12.2 %)
	≥ 3 ACEs 117 (13.8 %)
Age, Years, Mean (SD)	14.6 (1.6)
Number of ACEs, Mean (SD)	1.1 (1.8)

Table 2
Adverse Childhood Experience (ACEs) among students according to ACE-Q.

Items	Values
Emotional abuse	284 (21.9 %)
Physical abuse	129 (10.0 %)
Sexual abuse	11 (0.9 %)
Emotional neglect	186 (14.4 %)
Physical neglect	4 (0.3 %)
Parental divorce	143 (11.0 %)
Household physical violence	32 (2.5 %)
Household substance abuse	23 (1.8 %)
Household mental illness	41 (3.2 %)
Incarcerated household member	31 (2.4 %)
Number of ACEs	
0	434
1	192
2	103
≥ 3	117

Abbrev: ACE: adverse childhood experiences.

Table 3
Mental health disorders, information source and coping strategies among ethnic minority students.

Characteristics		Values
Stress	No	637 (75.3 %)
	Mild	116 (13.7 %)
	Moderate	74 (8.8 %)
	Severe	14 (1.7 %)
	Extremely severe	5 (0.6 %)
	Mean (SD)	5.68 (3.05)
Anxiety	No	386 (45.6 %)
	Mild	214 (25.3 %)
	Moderate	125 (14.8 %)
	Severe	62 (7.3 %)
	Extremely severe	59 (7.0 %)
	Mean (SD)	4.34 (3.01)
Depression	No	347 (41.0 %)
	Mild	209 (24.7 %)
	Moderate	226 (26.7 %)
	Severe	50 (5.9 %)
	Extremely severe	14 (1.7 %)
	Mean (SD)	5.62 (3.20)
Coping strategies	Reading books	353 (41.8 %)
	Listening to music	108 (12.8 %)
	Watching television	72 (8.5 %)
	Playing sport	90 (10.7 %)
	Sleeping	91 (10.8 %)
	Talking	48 (5.7 %)
	Crying	36 (4.3 %)
	Walking	47 (5.6 %)
	Mean (SD)	4.34 (3.01)
Information sources	Family	67 (7.9 %)
	School	300 (35.5 %)
	Friends	110 (13.0 %)
	Commune staff	42 (5.0 %)
	Youth Union	22 (2.6 %)
	Women Union	19 (2.3 %)
	Television	128 (15.2 %)
	Self-study	157 (18.6 %)
	Mean (SD)	4.34 (3.01)

using the Internet had a significantly lower anxiety score than those using the Internet at the weekend. Finally, information sources such as friends or television were negatively associated with stress and anxiety scores.

4. Discussion

Our study contributed to the pictures about mental health and ACEs in ethnic minority adolescents living in remote areas of Vietnam. The results of this investigation demonstrate a robust and affirmative correlation between ACEs and mental disorders in the context of ethnic minorities. There was an observed upward trajectory in the correlation

between the number of ACEs and the severity of mental disorders. Moreover, this study found several factors associated with mental disorders in this population, which could be used for developing further interventions.

In this study, we observed that 48.9 % experienced at least one ACEs during their lifetime. The rates of ACEs identified in our research exceed the mean occurrence rate observed among adolescents residing in low and middle-income nations (Blum et al., 2019) and significantly surpasses the prevalence documented in prosperous countries (Hughes et al., 2019). The current prevalence of ACEs is comparatively lower than that observed in a previous study conducted in 2020 among 4957 adolescents in Vietnam, which found that 86 % had at least ACEs (Thai et al., 2020). This could be attributed to the fact that the students participating in our study were all at ethnic boarding schools. Therefore, they rarely went back home and experienced fewer ACEs compared to adolescents in other studies in different settings. Despite this, the prevalence of ACEs in our study remained high. Our research findings reveal a notable rate of ACEs, which can be attributed to the limited knowledge and inadequate recognition of ACEs among adults in Asian nations. This may be attributed to the misconception prevailing among adults in such regions regarding what constitutes inappropriate treatment of adolescents (Dunne et al., 2008). Similarly, this phenomenon persists in Vietnam, where adults are less inclined to acknowledge emotional and physical maltreatment inflicted upon children as forms of child abuse. Instead, disciplinary measures and coercive strategies are employed to assert control and administer punishment. In developing countries and Asian nations, stakeholders commonly exhibit insufficient recognition of ACEs as a significant public health concern (Dunne et al., 2008). The persistent prevalence of ACEs and the persisting problem of their resolution in Vietnam and other Asian countries may be attributed to the traditional inclination to avoid discussing ACEs, hesitancy in intervening in the familial affairs of others, and the absence of a comprehensive reporting system.

In our study, we found a notably high rate of mental disorders among ethnic minority adolescents. Specifically, the rates of stress, anxiety and depression were 24.7 %, 54.4 % and 59.0 %, respectively. These rates were slightly lower than findings in a previous study in Hue, which found that 49.2 %, 61.5 % and 61.5 % of adolescents suffered from stress, anxiety and depression, respectively (Tran et al., 2023). The outcomes of this research hold significant significance, as the prevalence of self-reported data serves as an alarming signal of an underlying matter that necessitates immediate addressing. Considering the elevated prevalence of mental health issues amongst ethnic minority adolescents, it is recommended that early screening, diagnosis, and treatment be incorporated into standard health check-up programs. In addition, the provision of counselling support within educational institutions represents a crucial preliminary measure, especially in light of the limited comprehension surrounding the mental well-being of students.

The correlation between ACEs and symptoms of mental health has been consistently documented in a significant volume of global research literature. However, the strength of this correlation varies across different environments and populations (Hughes et al., 2017). The findings of our study suggest that Vietnamese ethnic minority adolescents who have experienced ACEs were more likely to have mental problems such as anxiety, stress or depression. Moreover, these results underscore the pressing necessity for interventions targeting ACEs in this population. The high prevalence of depression, anxiety and stress among Vietnamese adolescents could be attributed to a combination of ACEs, limited mental health literacy, and a preference for seeking help, as compared to their counterparts in developed nations (Thai et al., 2020). Our research indicates that students who utilise information sources such as community staff and television were associated with a reduced likelihood of experiencing mental health issues. Previous research conducted in Australia indicated that adolescents who utilised the Internet as a potential source were more likely to seek assistance for mental health issues (Reavley et al., 2011). Currently, Internet

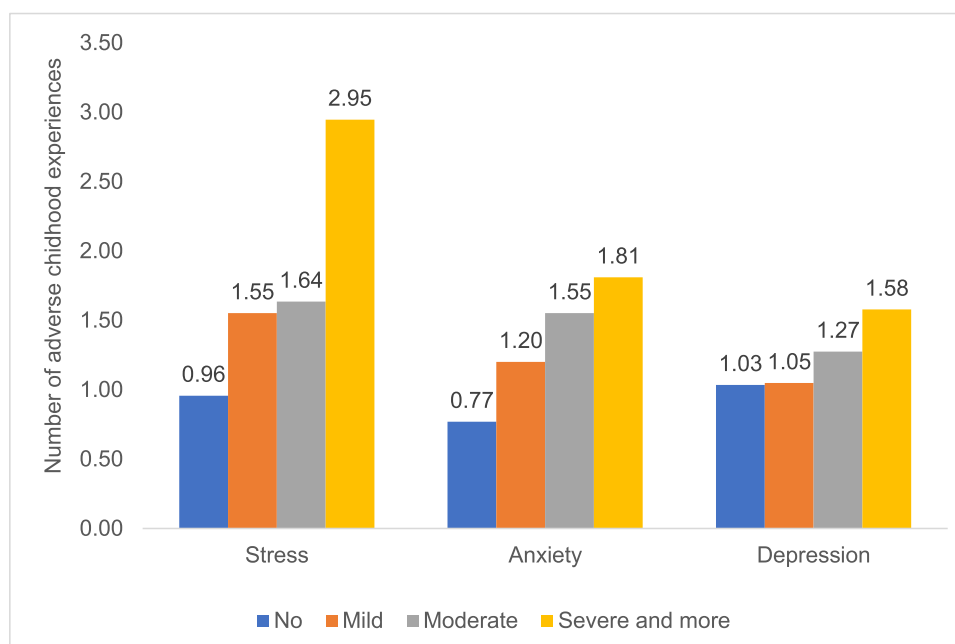


Fig. 1. Average number of adverse childhood experiences according to the severity of different mental disorders.

connectivity in mountainous provinces is not optimal, and there is a lack of guidance to help students effectively utilise online resources. The primary information from community staff or television is generally considered more reliable and official. Therefore, students who used these sources of information could have better mental well-being than other groups. It is noteworthy that we did not find any correlation between the source of information from the school and the mental health status of students. A previous study demonstrated the significant role of school-based information in enhancing adolescent mental health and promoting the utilisation of mental health counselling services (Green et al., 2013). This could signify that the role of schools in nurturing mental health may not be sufficiently prominent and requires further improvement.

The research findings also emphasise the role of friendship in reducing the risk of mental disorders among adolescent members of ethnic minorities compared to support from the family. A previous study also indicated that the level of closeness in friendships during mid-adolescence was predictive of subsequent increases in self-worth and reductions in anxiety and depressive symptoms during early adulthood (Narr et al., 2019). It also implied that adolescents who prioritise establishing strong interpersonal relationships are more proficient in navigating significant social developmental tasks than those prioritizing popularity among a wider peer group (Narr et al., 2019). The results of the regression model also indicate that information source from friends was a protective factor against mental health issues. Moreover, a higher number of friends the students lived with was negatively associated with mental problems such as anxiety. In a context where these adolescents have to go away from home to boarding schools, having friendships in both quantity and quality aspects plays an important role. Therefore, interventions to improve children's mental health can be based on this relationship to increase the feasibility of the intervention.

Several implications should be taken into consideration based on our research findings. The observed high prevalence of depression, anxiety, stress and ACEs underscores the imperative for intervention strategies targeting not only ACEs but also the associated mental health conditions. Based on the extensive linkage established between ACEs and mental disorders, it is imperative to acknowledge that addressing ACEs alone may not be deemed adequate. This is due to the presence of additional noteworthy factors influencing depression and psychological distress, which vary in significance across distinct contexts, particularly

in ethnic minority adolescents who had to study in boarding schools. Moreover, it is important to acknowledge that various forms of ACEs often intersect, necessitating the adoption of multidimensional approaches. For instance, instances of emotional abuse, physical abuse, and sexual abuse are collectively classified as ACEs; however, the strategies for intervention in each case may vary significantly, which could be based on friendships.

The present study is subject to certain limitations. One limitation of the study is the exclusive reliance on self-administered formats for data collection. This approach potentially introduces recall bias as it relies on individuals retrospectively recalling information. Additionally, the fear of punishment may lead to the under-reporting of adverse behaviours, further compromising the accuracy and comprehensiveness of the data collected. Moreover, the cross-sectional methodology employed in this study hinders the interpretation of causal relationships between explanatory factors and outcomes.

5. Conclusion

This study revealed the high prevalence of mental health problems and ACEs among ethnic minority adolescents in current sample. There is an urgent need for interventions to address ACEs and cognitive issues to improve adolescents' well-being.

CRedit authorship contribution statement

Ngo Anh Vinh: Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Vu Thi My Hanh:** Methodology, Project administration, Software, Writing – original draft, Writing – review & editing. **Do Thi Bich Van:** Investigation, Project administration, Software, Validation, Writing – original draft, Writing – review & editing. **Duong Anh Tai:** Formal analysis, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Do Minh Loan:** Investigation, Project administration, Validation, Writing – original draft, Writing – review & editing. **Le Thi Thanh Thuy:** Data curation, Software, Supervision, Writing – original draft, Writing – review & editing.

Table 4
Factors associated with anxiety, depression and stress in minority ethnic students.

VARIABLES	Anxiety (0=No, 1=Yes)		Anxiety score		Stress (0=No, 1=Yes)		Stress score		Depression (0=No, 1=Yes)		Depression score	
	OR	95 % CI	Coef.	95 %CI	OR	95 % CI	Coef.	95 %CI	OR	95 %CI	Coef.	95 %CI
Age											−0.11	−0.25; 0.03
Gender (Male-ref vs Female)					1.29	0.89; 1.85					0.40	−0.08; 0.88
Ethnicity (vs. Tay)												
Nung	0.85	0.62; 1.16										
Others	0.64*	0.38; 1.05										
Frequency of parental visit (vs.Once a month)												
Every two months	1.03	0.63; 1.70										
Others	0.34*	0.11; 1.09										
Number of people living together (<10 people-ref vs ≥ 10 people)	0.51***	0.33; 0.77	−0.87***	−1.44; −0.30	0.80	0.57; 1.12	−0.31	−0.73; 0.11	0.81	0.61; 1.07	−0.49*	−1.05; 0.07
Physical activity (Yes-ref vs No)	0.72**	0.51; 1.00										
Frequency of Internet use (vs. Weekend)												
No use	0.59	0.24; 1.47	−1.77***	−3.00; −0.53							−0.77	−2.08; 0.55
4 h/day	0.66	0.40; 1.10	−0.72**	−1.40; −0.04							−0.26	−0.91; 0.39
5 h/day	0.43**	0.21; 0.86	−1.29***	−2.24; −0.33							−1.35***	−2.33; −0.36
6 h/day or more	0.45**	0.22; 0.92	−1.26**	−2.25; −0.26							−0.07	−1.08; 0.93
Quality of friendships (vs. Good)												
Fair	1.34*	1.00; 1.81	0.54**	0.12; 0.96			0.47**	0.06; 0.89	1.30*	0.98; 1.73	0.55**	0.10; 1.00
Poor	1.77	0.75; 4.16	1.46**	0.31; 2.62			0.85	−0.29; 2.00	6.84***	2.03; 23.02	2.20***	0.96; 3.43
Information source (vs. Family)												
School	0.98	0.55; 1.75	0.04	−0.75; 0.84	0.75	0.42; 1.32	−0.46	−1.25; 0.33				
Friends	0.57*	0.30; 1.10	−0.91*	−1.82; 0.01	0.51*	0.25; 1.01	−1.41***	−2.32; −0.50				
Commune staff	0.80	0.35; 1.83	−0.31	−1.49; 0.86	0.60	0.24; 1.49	−1.20**	−2.36; −0.03				
Youth Union	0.65	0.24; 1.82	−0.95	−2.40; 0.50	1.19	0.43; 3.30	−0.34	−1.78; 1.11				
Women Union	1.09	0.37; 3.23	−0.12	−1.65; 1.41	0.35	0.09; 1.35	−1.08	−2.61; 0.44				
Television	0.54*	0.28; 1.02	−1.31***	−2.21; −0.40	0.37***	0.18; 0.76	−1.42***	−2.31; −0.53				
Self-study	0.82	0.44; 1.54	−0.73	−1.61; 0.15	0.55*	0.29; 1.04	−0.79*	−1.65; 0.07				
Number of ACEs	1.20***	1.09; 1.33	0.28***	0.16; 0.40	1.20***	1.10; 1.30	0.28***	0.17; 0.40	1.07	0.98; 1.16	0.16**	0.03; 0.30

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Abbrev: ACE: adverse childhood experiences.

Declaration of competing interest

We confirm that we have no potential conflict of interest.

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